Basic I/O Operation – Performance (Computer Organization / Operating Systems)

Input/Output (I/O) performance refers to how efficiently a computer system transfers data between the **CPU, memory, and I/O devices** such as keyboard, disk, printer, and network devices. Efficient I/O operations are important because I/O devices are usually **much slower than the CPU**, which can cause delays in system performance.

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1. Basic I/O Operations

The basic steps involved in an I/O operation are:

1. **I/O Request**
   The CPU sends a command to the I/O device controller to start an operation.
2. **Device Controller Action**
   The device controller receives the command and performs the operation (reading or writing data).
3. **Data Transfer**
   Data is transferred between the device and the **main memory**.
4. **Completion Signal**
   After completing the operation, the controller sends an **interrupt signal** to the CPU.
5. **CPU Response**
   The CPU processes the interrupt and continues execution of the program.

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2. I/O Performance Metrics

Performance of I/O systems is measured using several parameters:

a) Throughput

- The amount of data transferred per unit time.
- Measured in **MB/s or GB/s**.
- Higher throughput means better I/O performance.

b) Response Time

- The time taken for a device to respond to an I/O request.
- Includes **waiting time + data transfer time**.

c) Latency

- The delay between sending a request and starting the data transfer.

d) Bandwidth

- The maximum rate at which data can be transferred.

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3. Factors Affecting I/O Performance

1. Device Speed

Different devices operate at different speeds.

Example:

- Keyboard – very slow
- Hard disk – medium
- SSD – very fast

2. Bus Speed

The system bus affects how quickly data moves between CPU, memory, and devices.

3. I/O Method Used

Different I/O techniques influence performance.

Programmed I/O

- CPU continuously checks the device status.
- CPU is busy waiting.
- Low performance.

Interrupt Driven I/O

- Device sends interrupt when ready.
- CPU performs other tasks meanwhile.
- Better performance.

Direct Memory Access (DMA)

- Data transfer occurs directly between device and memory.
- CPU involvement is minimal.
- Highest performance.

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4. Techniques to Improve I/O Performance

Buffering

Temporary memory used to store data during transfer.

Advantages:

- Reduces waiting time
- Smoothens data transfer speed differences

Caching

Frequently used data is stored in fast memory.

Advantages:

- Reduces access time
- Improves system efficiency

Spooling (Simultaneous Peripheral Operations On-Line)

Data is stored in disk before sending to a slow device like a printer.

Advantages:

- Allows multiple programs to share devices
- Improves overall performance

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5. Example of I/O Performance Improvement

Example: **Disk Read Operation**

1. Program requests disk data.
2. Disk controller reads data.
3. Using **DMA**, data moves directly to memory.
4. CPU continues executing other tasks.
5. Interrupt notifies CPU when transfer completes.

This reduces CPU idle time and improves performance.

Problems on Basic I/O Operation – Performance

Problem 1: I/O Data Transfer Time

A disk transfers data at a rate of **50 MB/s**.
If the system needs to transfer **200 MB** of data, calculate the **time required** for the transfer.

Solution

Formula:

$$Transfer\ Time\ = \frac{Data\ Size}{Transfer\ Rate}$$

Given:

- Data size = **200 MB**
- Transfer rate = **50 MB/s**

$$Transfer\ Time\ = \frac{200}{50}Transfer\ Time\ =\ 4\ seconds$$

Answer

The **data transfer time = 4 seconds**.

Problem 2: CPU Utilization in Programmed I/O

A device transfers **1 byte per 2 μs**.
The CPU requires **0.5 μs** to check device status.

Calculate the **CPU utilization during programmed I/O**.

Solution

Total time per transfer = **2 μs**

CPU busy time = **0.5 μs**

Formula:

$$CPU\ Utilization\ = \frac{CPU\ Busy\ Time}{Total\ Time}CPU\ Utilization\ = \frac{0.5}{2}CPU\ Utilization\ =\ 0.25CPU\ Utili$$

Answer

CPU utilization = **25%**

Problem 3: Throughput Calculation

A printer prints **20 pages per minute**.
Each page contains **500 KB of data**.

Find the **data throughput**.

Solution

Total data printed per minute:

$$20 \times 500KB = 10000KB 10000KB = 10MB$$

Throughput per second:

$$\frac{10MB}{60} = 0.167MB/s$$

Answer

Printer throughput = **0.167 MB/s**

Problem 4: DMA Transfer Time

A DMA controller transfers **4096 bytes** of data at a speed of **1 MB/s**.

Calculate the **time required for the transfer**.

Solution

Convert bytes to MB:

$$4096\, bytes = 4KB 4KB = 0.004MB$$

Formula:

$$Time = \frac{Data}{Rate} Time = \frac{0.004}{1} Time = 0.004\, seconds = 4\, ms$$

Answer

DMA transfer time = **4 milliseconds**

Problem 5: Disk I/O Performance

A disk has:

- Seek time = **8 ms**
- Rotational delay = **4 ms**
- Transfer time = **2 ms**

Find the **total disk access time**.

Solution

Formula:

$$Disk\ Access\ Time\ =\ Seek\ Time\ +\ Rotational\ Delay\ +\ Transfer\ Time\ =\ 8\ +\ 4\ +\ 2\ =$$

Answer

Total disk access time = **14 milliseconds**